

Lab 6: Map

Nguyễn Thanh Hùng – 2540056

Each 2D thread transforms one grayscale pixel: threshold 128 for binarization, clamped $1.2p + 30$ for brightness, or $0.5I_1 + 0.5I_2$ for blending. The second image is resized to match the first. Several 2D blocks are timed, and NumPy references validate the saved outputs.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.